

# Not all differences matter. So who decides which ones do?

Governed human judgment in agentic-AI variability exploration — a structured exploratory literature review

## RESEARCH QUESTION

*How can human judgment be captured, governed, and used to support agentic-AI-driven variability exploration in guideline operationalization scenarios, enabling reliable human–AI co-reasoning?*

## Ali Hamed

M.Sc. Information Systems · University of Haifa

Supervisors: Prof. Iris Reinhartz-Berger · Prof. Arnon Sturm

*Doctoral working draft · research-question wording provisional, not supervisor-approved · no empirical or medical claim.*

# An AI can find every difference — not which ones matter

The same structural deviation can be three different things, and only situated judgment separates them.



## A legitimate alternative

A different representation can be valid under the same language semantics and domain meaning. Penalising it teaches the wrong lesson.



## A genuine error

A misconception, omission, syntax violation or domain mistake that should be corrected.



## A defective guideline

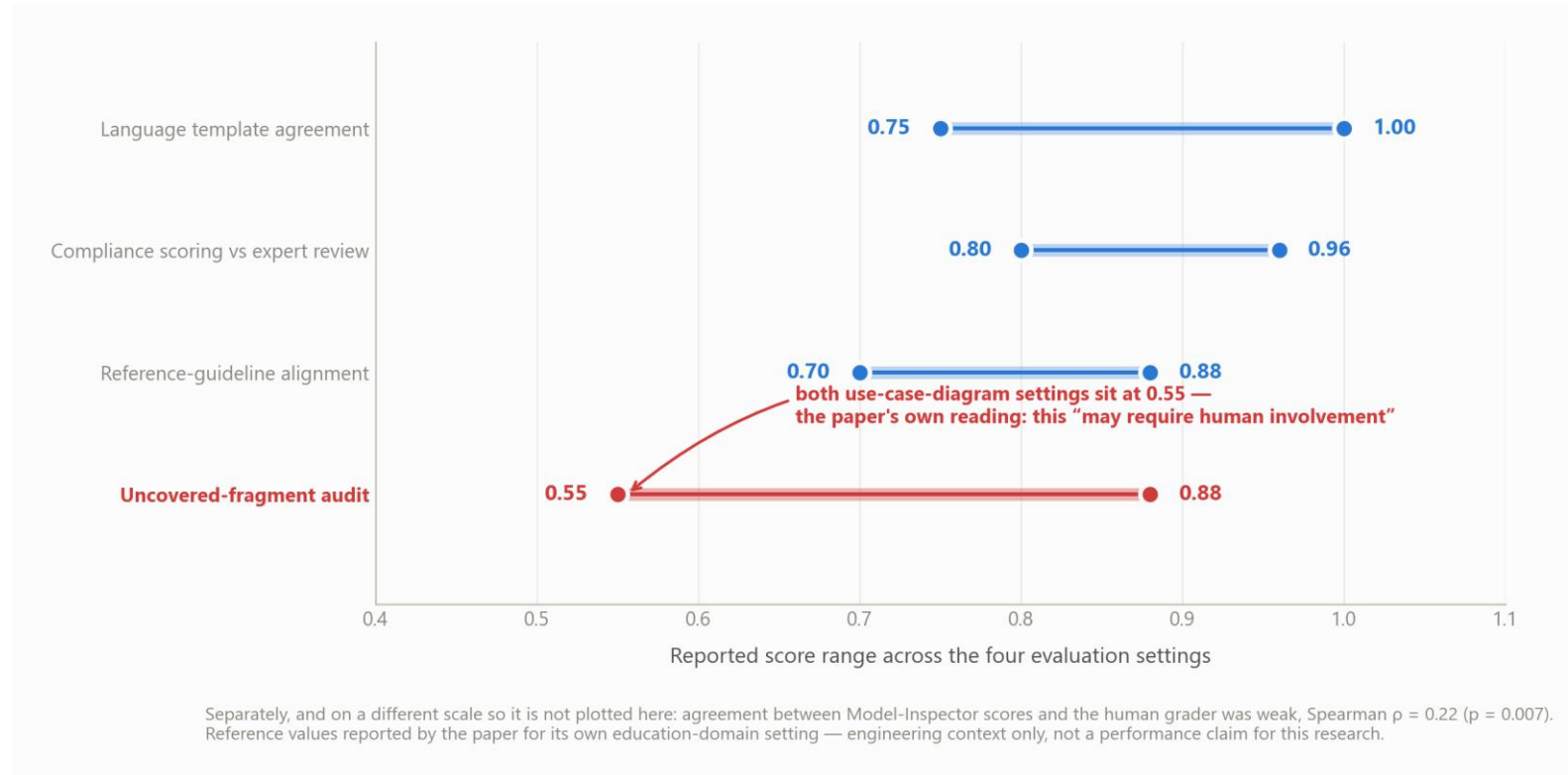
The reference itself may be incomplete, underspecified or mismatched to the case — the deviation is evidence against the norm.

**“Human-in-the-loop” is not one problem. It is four design problems:**

when to ask · what to capture · how to govern · when to reuse

# The framework names its own need for a human

VEGO-AI assesses domain models with four coordinated LLM agents. It works — unevenly, and it says so.



$\rho = 0.22$

Reported correlation between Model-Inspector compliance scores and the human grader.

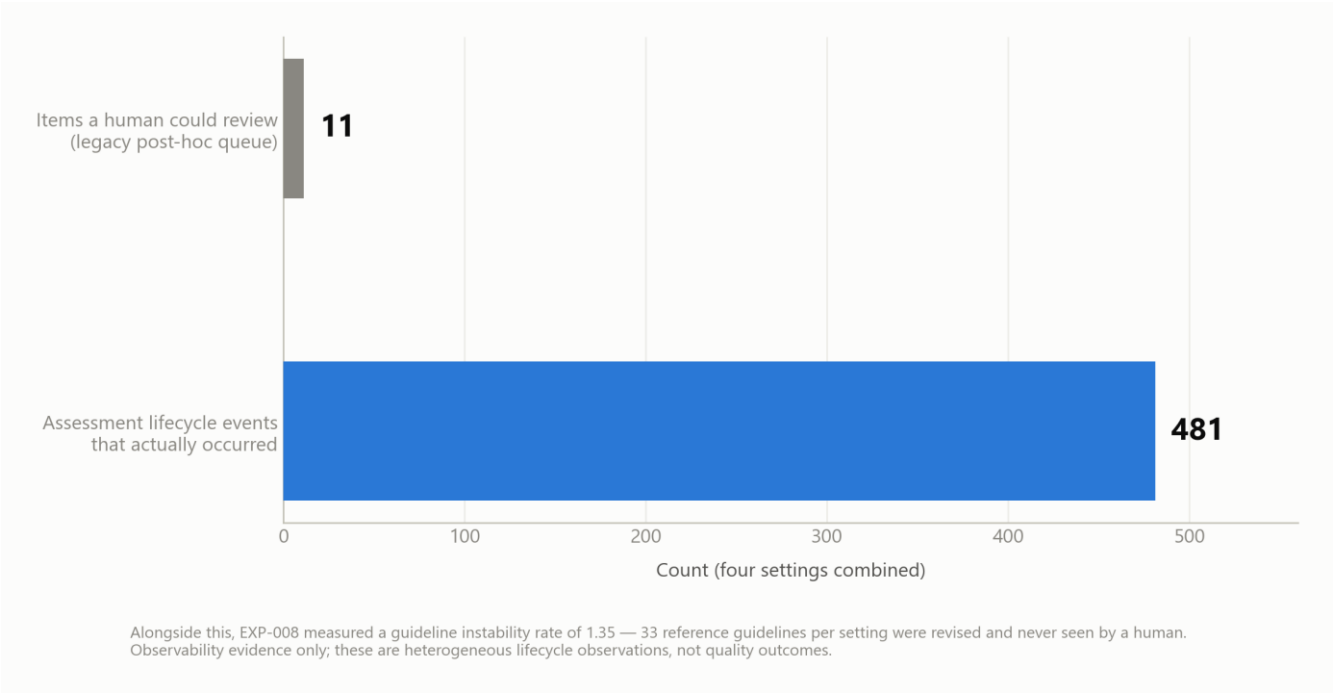
**The paper's own future work:**

*"incorporating human-in-the-loop oversight at key pipeline stages (guideline validation, compliance review, and variability classification) would strengthen reliability"*

**The need for human judgment is not my assumption. It is the framework's own stated next step — and it names the three stages.**

# The same gap is measurable inside the pipeline

Replaying the assessment pipeline as an event stream shows how little of its reasoning was ever reviewable — and that asking more often does not fix it.



44×

more lifecycle events occurred than the legacy review queue could show

33

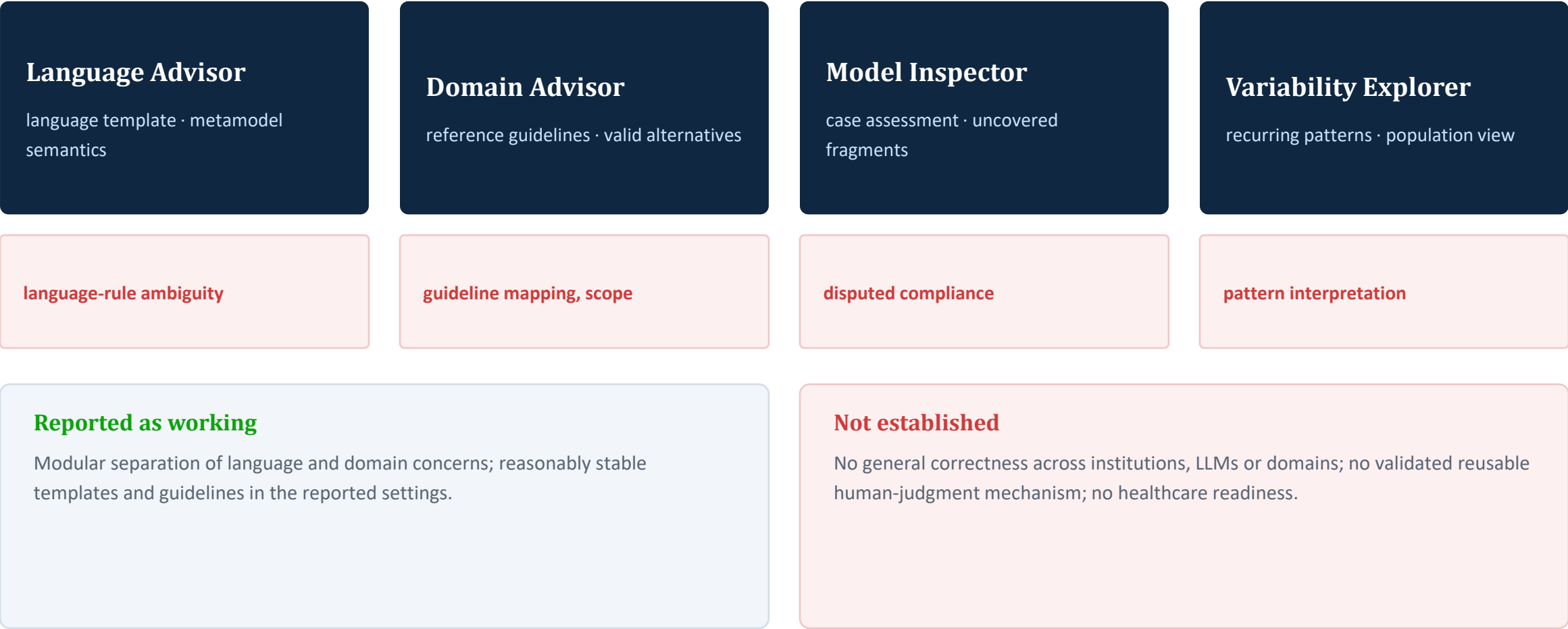
reference guidelines per setting revised with no human ever seeing them

0.80+

share of events a human must see to reach 80% high-severity coverage

# Where judgment would enter the pipeline

Four coordinated agents. The review asks what a governed human-judgment layer around them would have to do.



# The research question

*How can human judgment be captured, governed, and used to support agentic-AI-driven variability exploration in guideline operationalization scenarios, enabling reliable human–AI co-reasoning?*

## **It names the task, not the solution**

The object of study is variability exploration in guideline operationalization — not a particular agent architecture.

## **It is narrower than “human–AI collaboration”**

Intervention and reuse must attach to a real decision unit: a guideline mapping, a fragment classification, a pattern.

## **Reuse is not assumed in the headline**

Whether captured judgment may be reused is a governed, testable concern for the sub-questions — not a premise.

# Three derived questions — which become three studies

Each sub-question owns one stage: when to ask, what to keep and govern, and where a ruling may travel.

SQ1

## When to ask

When and how should an agentic assessment system request human judgment, so that important uncertainties are addressed without unnecessary expert burden?

Study 1 · selective intervention

SQ2

## What to capture and govern

How should expert judgment — including the system's own reasoning — be represented, validated, reconciled and stored so it can be reused transparently, without unsafe generalization or loss of human authority?

Study 2 · judgment lifecycle

SQ3

## When to reuse and transfer

How can expert judgment be reused and transferred across guideline-operationalization contexts — what generalizes, and what must adapt?

Study 3 · transfer and evaluation

Separating the studies prevents one small label set from being treated as evidence for intervention policy, representation validity and transfer at once. The umbrella question is answered only after they integrate.

# How the review was done

A structured exploratory review with systematic controls — organised by concepts, not by paper summaries.

## Relevant research areas

**Human involvement in agentic AI — the core** mixed initiative, levels of agency, oversight, control

**Selective intervention** learning to defer, abstention, uncertainty, value of information, burden

**Judgment capture and governance** reasoning capture, provenance, disagreement, authority, expiry, revocation

**Memory, reuse and transfer** scope-aware retrieval, domain and guideline adaptation, leakage-safe evaluation

*Deliberately outside the review body: enabling technologies (model architectures, local-inference tooling). They are engineering means — the review must justify the problem, not describe the solution.*

## Screening

**Include** methods, frameworks, artifacts, empirical studies.

**Exclude** duplicates, passing mentions, off-topic enabling tech, unverifiable sources.

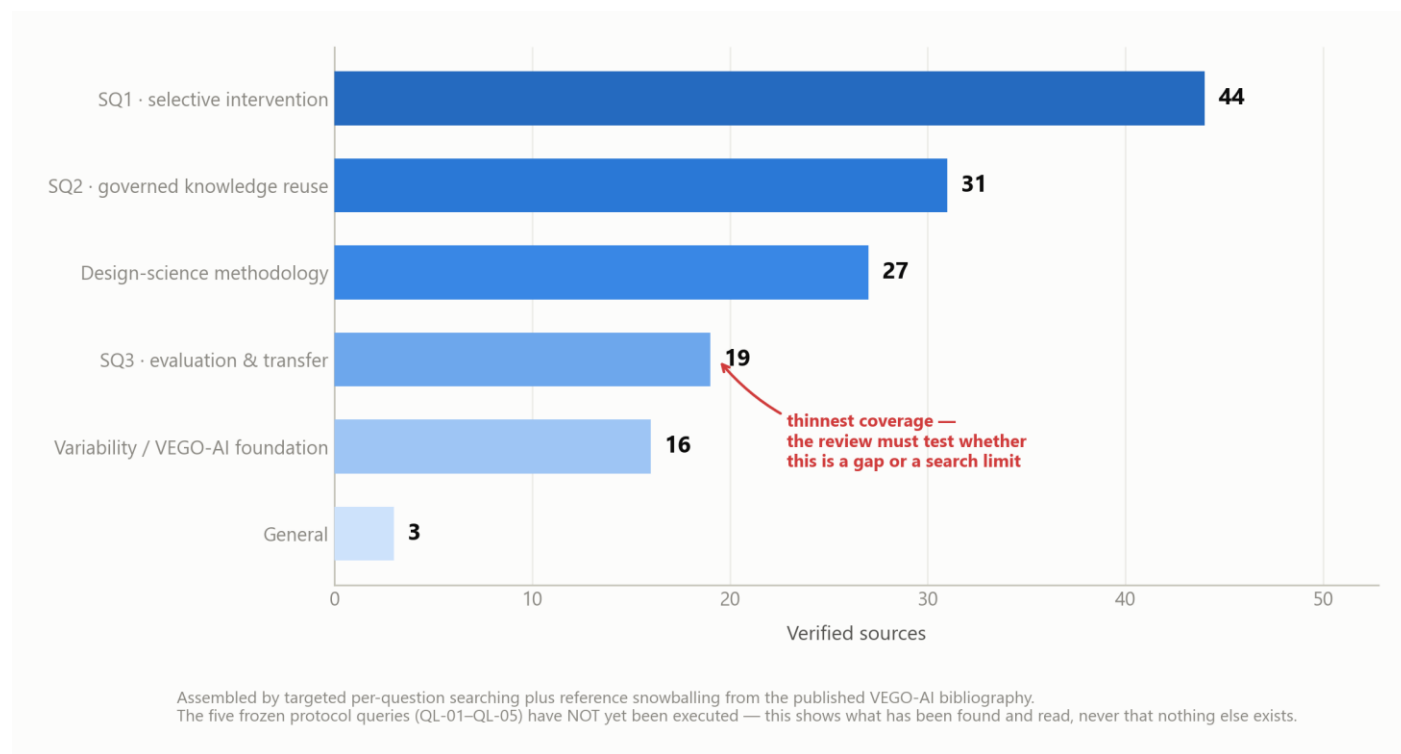
## GenAI disclosure

Used for organisation, synthesis drafting, diagrams and consistency checks. Not used to approve inclusion, invent citations, or create expert labels.



# Search strategy — and what has actually been executed

The query families are locked. Presenting them is not the same as claiming the formal searches have been run.



## Locked query family — SQ1

(learning to defer OR algorithmic triage OR abstention OR selective prediction) AND (human OR expert) AND (cost OR burden OR timing OR uncertainty OR disagreement)

Sources: Scopus · ACM DL · IEEE Xplore · SpringerLink · Web of Science. Scholar and dblp for discovery, snowballing and verification.

## Execution status

QL-01–QL-05: **0 of 5 executed**

20 RQ-anchor mappings verified (5 per question); evidence maturity 14 full-text + 6 record-level. This is what has been found and read — not evidence that nothing else exists.

# Six streams — each governs part of the lifecycle

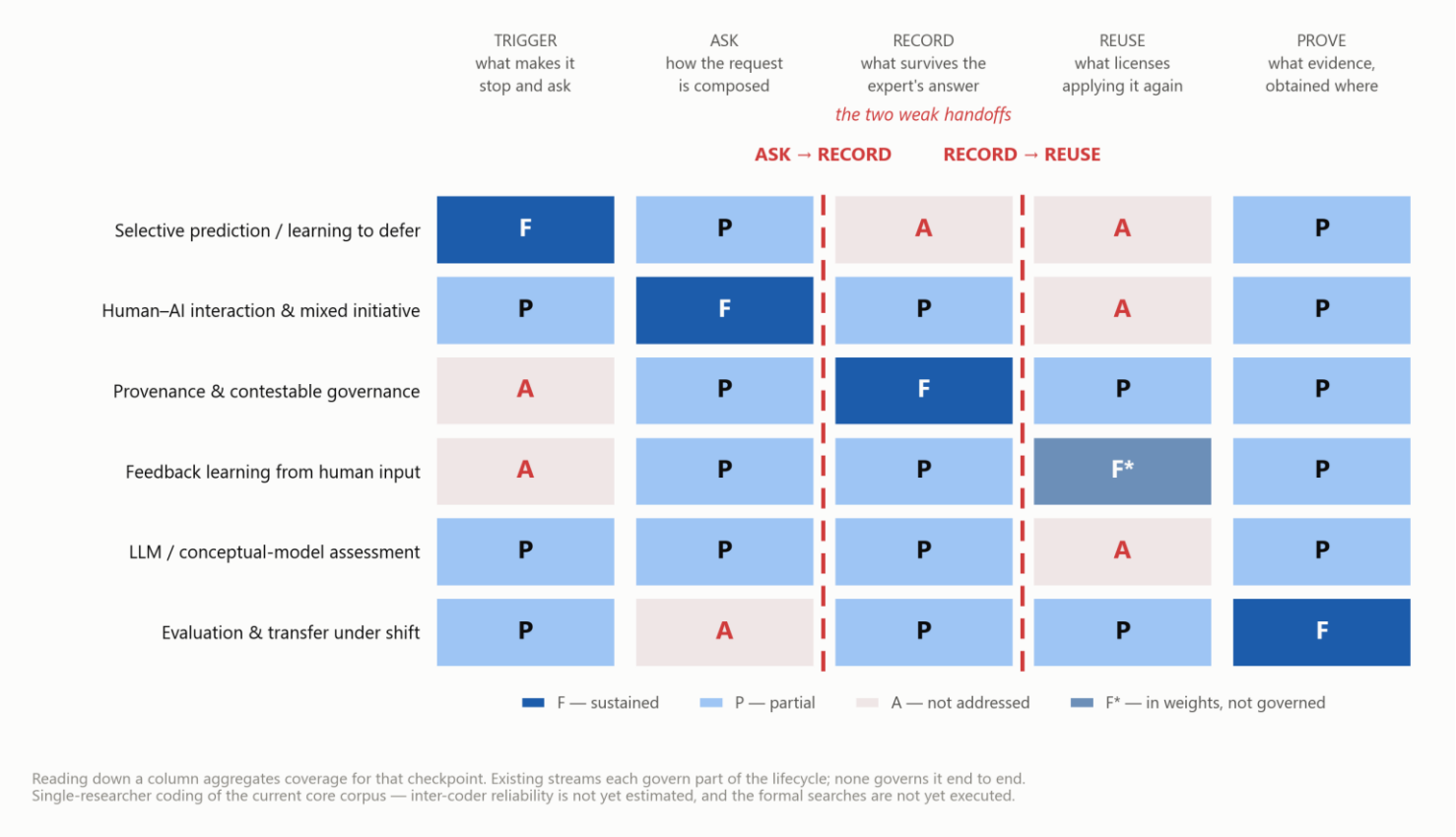
Concept matrix from the review: what each stream asks, and what it leaves open for governed judgment in agentic assessment.

|                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>1</div><div>Learning to defer / reject option <span>SQ1</span></div><div>When to route a decision to a human — uncertainty, expert performance, cost, selective risk.</div><div>Leaves open usually treats the expert's response as a terminal label.</div></div> | <div><div>2</div><div>Mixed initiative &amp; human-centred AI <span>SQ1</span></div><div>Who acts and who controls — initiative timing, transparency, correction, override.</div><div>Leaves open does not specify a reusable judgment object.</div></div>              | <div><div>3</div><div>Explanatory debugging &amp; disagreement <span>SQ2</span></div><div>How users correct reasoning, and how multiple human judgments relate — dissent, ambiguity, reconciliation.</div><div>Leaves open rarely addresses long-term scope and revocation.</div></div> |
| <div><div>4</div><div>Provenance &amp; documentation <span>SQ2</span></div><div>Where evidence and decisions came from — lineage, attribution, purpose, limitations.</div><div>Leaves open does not decide when to ask, or when reuse is permitted.</div></div>             | <div><div>5</div><div>Retrieval &amp; external memory <span>SQ3</span></div><div>How prior evidence is retrieved — non-parametric access over versionable sources.</div><div>Leaves open similarity alone grants neither authority nor transfer permission.</div></div> | <div><div>6</div><div>Domain &amp; guideline adaptation <span>SQ3</span></div><div>How knowledge moves across contexts — source–target comparison, local validation, transparency.</div><div>Leaves open not tailored to granular human judgments inside agent pipelines.</div></div>   |

Concept matrix condensed from structured review v9 §11.1. Each “leaves open” is a residual limitation of the reviewed work — not a claim that the field contains nothing further.

# Judgment-lifecycle coverage across the streams

The unit of analysis is not a paper and not a topic — it is one expert ruling, and the five checkpoints it must pass from the moment an agent hesitates to the moment someone claims the ruling still holds elsewhere.



WHAT THE COLUMNS REVEAL

ASK → RECORD

Streams that are strong at eliciting a judgment are weak at keeping it. The correction is captured, then the reasoning behind it is discarded.

RECORD → REUSE

Streams that are strong at storing are weak at licensing. Retrieval fires on similarity; nothing checks whether this ruling may govern this new case.

No stream spans the loop

Each governs part of the lifecycle. The contribution opportunity is the connection between them, not a missing isolated algorithm.

# What the framework makes visible

Each gap is a thin or empty region of the matrix, paired with the artifact hypothesised to close it.

Six residual gaps — each with a matching contribution hypothesis

|    |                                                   |                                                                                                                                                                                                                                |                                                  |
|----|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| G1 | Bounded-attention intervention                    | Deferral and active-learning work supplies routing methods, but not a policy combining uncertainty, consequence, disagreement, novelty, reuse value and queue state at fragment and pattern level.                             | C1 · Selective Human Review Orchestrator         |
| G2 | Reasoning-rich judgment representation            | Feedback systems capture corrections and annotation systems capture labels; the joint representation of decision trace, rationale, evidence, scope, authority and outcome is not established for agentic modelling assessment. | C2 · Governed Judgment Object                    |
| G3 | Governance lifecycle                              | Provenance and contestability are well motivated, but operational rules for validation tiers, conflicts, supersession, expiry, revocation and claim-specific authority are missing.                                            | C3 · Contestable Judgment Store                  |
| G4 | Scope-aware memory                                | Retrieval augmentation provides retrieval, but not permission. The missing mechanism combines semantic retrieval with hard governance gates and advisory use.                                                                  | C4 · Scope-Aware Retrieval Advisor               |
| G5 | Transfer and leakage controls                     | Adaptation research motivates local validation, but the transfer properties of individual expert judgments and the correct frozen-store evaluation design remain open.                                                         | C5 + C6 · Transfer Ladder · Leakage-Safe Harness |
| G6 | Construct validity of the variability distinction | “Substantial” versus “occasional” mixes validity with frequency language, and may obscure guideline mismatch, ambiguity and distinct error types.                                                                              | Terminology / rubric revision                    |

C1–C6 are design and research-contribution hypotheses. They may be called implemented only when the artifact exists and passes tests, and empirically beneficial only after independent evaluation.

# The gap is integration — and here is what would falsify it

## The gap is integration, not an isolated missing algorithm.

Mixed initiative already manages initiative; deferral already allocates cases; provenance already encodes lineage; retrieval already surfaces evidence; adaptation already models shift. What is unresolved is how to connect them into one traceable loop: request judgment selectively, capture it at the reasoning level, govern it as a versioned lifecycle object, reuse it only as scope-filtered advice, and evaluate it across a frozen leakage boundary.

1

### Finalize RQ terminology

supervisor decision log; D-RQ-01/02

2

### Execute the locked searches

QL-01–QL-05, currently 0 of 5

3

### Collect independent expert labels

EXP-005 at 0/24; ≥20 targeted

4

### Evaluate SQ1 and SQ2 first

burden, traceability, same-domain reuse

5

### Then transfer, frozen-store

matched control + blind target labels

6

### Medical only after gates

readiness 0/6; Plan B stays software

Human judgment should be requested selectively, captured with its reasoning, governed as a contestable lifecycle object, reused only as scoped advice, and evaluated across a frozen leakage boundary.

Thank you — questions welcome.

# Exact provisional wording and construct boundaries

Preserved verbatim so the wording can be challenged precisely.

## Provisional questions

**U-RQ** How can human judgment be captured, governed, and used to support agentic-AI-driven variability exploration in guideline operationalization scenarios, enabling reliable human-AI co-reasoning?

**SQ1** When and how should an agentic assessment system request human judgment, so that important uncertainties are addressed without unnecessary expert burden?

**SQ2** How should expert judgment — including the system's own reasoning — be represented, validated, reconciled and stored so it can be reused transparently, without unsafe generalization or loss of human authority?

**SQ3** How can expert judgment be reused and transferred across guideline-operationalization contexts — what generalizes, and what must adapt?

## Construct boundaries

**Human judgment** a context-bound assessment, correction, rationale or policy decision by an authorized person — not a label.

**Expert judgment** authority tied to domain, language, pedagogical or governance competence; partial and contestable.

**Substantial variability** a contextually justified alternative consistent with language semantics and domain logic.

**Occasional variability** an error, misconception or non-compliance — the term does not mean rare, which is a construct-validity problem flagged as G6.

**Reusable judgment** a governed record retrievable under explicit scope, authority, version and validation conditions.

# Evidence levels and permitted claim language

Every statement in the review is bounded by the maturity of the evidence behind it.

|           |                                                                                                                      |                                                                 |
|-----------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| FT-A      | Official or author-hosted full text reviewed for method, finding and limitation. Current count: 14.                  | May support bounded detailed claims with a locator.             |
| FT-B      | Substantial authoritative excerpt or repository record reviewed; full article not fully inspected. Current count: 6. | May support only the directly visible claim.                    |
| ID-S      | Bibliographic identity verified — title, venue, DOI or authoritative record checked.                                 | Discovery and candidate relevance only.                         |
| AB-S      | Title or abstract screened without sufficient full-text evidence.                                                    | Search-space description only — never gap proof.                |
| Synthesis | Reasoning that integrates sources or proposes a mechanism.                                                           | Must be labelled synthesis, implication or research hypothesis. |

# SQ1 — selective intervention is a resource-allocation problem

Uncertainty alone is insufficient. Priority must also weigh consequence, novelty, disagreement, reuse value and burden.

## Trigger signals

- Calibrated uncertainty
- Cross-agent disagreement
- Novelty or coverage gap
- Consequence and policy importance
- Weak underlying evidence
- Expected value of information
- Reviewer burden and queue state

## Intervention modes

**Proceed and log** low impact, familiar

**Focused review** uncertain but bounded

**Queue or batch review** important, not urgent

**Interrupt or stop** high consequence

*Provisional answer: request judgment when the expected value of expert information exceeds review burden and interruption risk — targeting the smallest decision unit.*



# SQ2 — a reusable judgment is not a naked label

The minimum reusable unit carries its grounding, its reasoning, its authority and its lifecycle.

1

## Case grounding

artifact · fragment · guideline version · evidence locator

2

## Decision trace

system claim · confidence · concise reasoning · correction

3

## Governance

authority · scope · visibility · dissent · permission

4

## Lifecycle

validated · expired · superseded · revoked · active

5

## Reuse signals

semantic signature · counterexamples · thresholds

6

## Outcome trace

retrieval · influence · override · downstream effect

## Reasoning-capture rule

Store the shortest sufficient decision trace — claim, evidence, rule, uncertainty and correction. Not hidden chain-of-thought, and not unnecessary personal data.

# SQ3 — retrieval is not permission

Reuse is advisory and scope-filtered; evidence requirements rise with transfer distance.

|    |                                  |                                               |
|----|----------------------------------|-----------------------------------------------|
| L0 | <b>same-case retry</b>           | default relevance high                        |
| L1 | <b>same task and domain</b>      | advisory after scope checks                   |
| L2 | <b>new task, same domain</b>     | independent labels + holdout                  |
| L3 | <b>adjacent domain</b>           | preregistered restricted experiment           |
| L4 | <b>cross-domain / healthcare</b> | no default permission; ethics + authorization |

## Leakage-safe evaluation

- Frozen source store and retrieval policy
- Matched no-reuse control
- Independent blind target labels, set before source judgments
- Store updated only after scoring
- Predeclared thresholds

# Three studies, each independently falsifiable

Separate units of analysis prevent one small label set from carrying every claim.

|                   |                                                     |                                                           |                                                                                     |
|-------------------|-----------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------|
| Study 1 / SQ1     | Selective Human Review Orchestrator                 | Held-out cases, policy baselines, burden logging          | Falsified by: no gain over a simpler baseline, or unacceptable burden               |
| Study 2 / SQ2     | Governed Judgment Object + Contestable Store        | Label-only comparator, reconstruction and lifecycle tests | Falsified by: reviewers cannot use the object, or governance cost outweighs benefit |
| Study 3 / SQ3     | Scope-Aware Retrieval Advisor + transfer classifier | Frozen store, blind labels, matched no-reuse control      | Falsified by: no target benefit, or effect vanishes under blinding                  |
| Integrated / U-RQ | End-to-end governed lifecycle                       | Human-only, AI-only and non-governed baselines            | Falsified by: any load-bearing study fails, or gain is only extra human effort      |

# Design science, GenAI use, and the evidence boundary

How this sits in the course's design-science frame, and exactly what is not being claimed.

## Design science

**Relevance** expert review is scarce; deviation meaning is contextual; guidelines drift unseen

**Design** selective-intervention policy + governed judgment lifecycle, built and evaluated per sub-question

**Rigor** structured review with evidence levels; the lifecycle framework is itself a candidate contribution back

## GenAI — used for

- Literature organisation
- Synthesis drafting
- Diagram production
- Consistency checks
- Document layout

## Not used for

- Approving inclusion decisions
- Inventing citations
- Creating expert labels
- Claiming supervisor approval

## Not claimed

- No improvement in assessment accuracy — expert labels stand at 0 of 24.
- No reduction in expert workload.
- No safe generalization or transfer.
- Nothing about clinical performance — medical readiness 0 of 6.
- No claim that the literature holds no prior solution — the formal searches are not yet executed.